What was our last week Discussion?

1. Strings & lists

```
s = "Programming"
#what is the output
s[::-1]

s = "Programming"
# what is the output
s[::-1]

{"type":"string"}

fruits = ["apple", "banana", "mango"]
#how do we check if 'apple' is in the list

fruits = ["apple", "banana", "mango"]
# we use in
# {what we want} in {where we are looking for}
"apple" in fruits

True

name = "Zechariah"
"a" in name

True

'''
write a program to check for palindrome (is a word that's read the
same from left to right and
right to left, for example like: madam, level, racecar'''


#hint: reverse the string first, then compare
word = "level"
reverse = word[::-1]

word == reverse #comparing our original word and the reversed one


True

# let's ask a user for a word
word = input("Please give me a word to check for palindrome: ")
reverse = word[::-1]
#let's check if it is palindrome
word == reverse

Please give me a word to check for palindrome: madam
```

```
True

#ask a user for a word
word = input("Please insert any word: ") #madrid .. level
#reverse teh word
reverse = word[::-1] #dirdam ... level
#check if the word is palindrome
#if the word is palindrome: print "the word is palindrome"
if word == reverse: # "madrid" == "dirdam" ... False ,... "level" ==
"level"...True
  print("the word is palindrome")
#else print "it is not palindrome"
else:
  print("it is not palindrome")

Please insert any word: level
the word is palindrome
```

2. control flow

Checking for multiples of a number

we use `%(modulo)` for checking if a number is multiple or factor

the logic is

if we have a number named "number"

we wanted to check if it is multiple of a number d

```
number % d == 0 #this is how we always check

number = 45
d = 2 # we wanted to check if 45 is multiple of 2
45 % 2 == 0

False

number = 498
d = 5 #we want to check if 498 is divisible by 5
number % d == 0

False

45 % 2

1

48 % 2

0
```

```
#ask user for a number
number = int(input("enter a number: "))
#chek if it's multiple of three
if number % 3 == 0:
  print("The number is multiple of three")
else:
  print("The number is not multiple of three")
```

```
enter a number: 58
The number is not multiple of three
```

```
# 50 is multiple of 5
# *, /, //. %, **

#check if 42 is multiple of 3

'''
    2.
        i. ask the user to insert a number n,
        ii. write conditional statment that will
            a. prints "FizzBuzz" if it is multiple of both 3 and 5
            b. print out "Fizz" if the number is multiple of 3
            c. prints "Buzz" if it is multiple of 5
            d. prints the number itself if it's neither.
 '''

n = int(input("Insert a number: "))



Insert a number: 78

'''
    3. we know temperature conversion from Fahrenheit to Celsius and
viseversa
    C = (5/9) * (F - 32) and F = (C * 9/5) + 32
    Given number with temperature value and letter with the temprature
type, convert it to the appropreiate one.
'''

'''
    4. Ask the user to enter thier grade out of 100.
    the criteria is, if the grade is above 90, it's an A,
    between 85 and 90 is B
    between 80 and 85 is C
    between 70 and 80 is D
    anything below 70 is F
'''
```

```
'''
    5. ask user to enter a number.
        if the number is positive, multiply it by 10
        if the number is negative, multiply is by 5
        if it's 0, add 20 on the number
'''

number = int(input("enter a number: "))
```

```
enter a number: 90
```

```
'''
    6. ask user to insert two numbers and an operation(either + or -).
then perform the operation.

    for example:
        enter the first number: 10
        enter the second number: 5
        enter operation to perform: +  --> this means we want to add
them
        so the result will be: 10 + 5 = 15
    example 2:
        enter the first number: 20
        enter the second number: 3
        enter operation to perform: -  --> this means we want to add
them
        so the result will be: 20 - 3 = 17

'''

'''
    7. ask user to insert a number from 1 to 5.
     based on this number print hte day of the
     week.
        1 means "Monday"
        2 means "Tuesday"
        3 means "Wednesday"
        4 means "Thursday"
        5 means "Friday"
'''

'''
    8. A company decided to give bonus of 5% to employee if his/her
year of service is more than 5 years.
    Ask user for their salary and year of service and print the net
bonus amount.
'''

# studnet should be from higa and his/her phy and math grade should be
above 90
```

```
school = input("Where do you study? ")
physics = float(input("what's your physics score? "))
math = float(input("what's your math score? "))


Where do you study? Higa
what's your physics score? 89
what's your math score? 90
sorry you are not accepted

school = input("Where do you study? ")
physics = float(input("what's your physics score? "))
math = float(input("what's your math score? "))

#do the same logic but using and instead of nested condition

Where do you study? Higa
what's your physics score? 90
what's your math score? 89
```

Introduction to Loops

```
oneToFive = [1, 2, 3, 4, 5 ]
oneToFive

[1, 2, 3, 4, 5]
```

Imagine we have list named primes

```
primes = [2, 3, 5, 7]
# if we want to print every item in the list what do we do?
# we print out every item one by one like below
print(primes[0])
print(primes[1])
print(primes[2])
print(primes[3])
```

But it is very tiresome job if we have list of size 100 for example.

In General when we have repeated job we use loops

For Loops

- **`For loops iterate over a given sequence`**

syntax

```
for varaible in sequence:
        do something
```

Remember: sequence means list, string...

```
primes = [2, 3, 5, 7]
for variable in primes: #variable takes values in primes one by one in each iteration
  print(variable) #2 3 5 7

#let's say we have list named primes
primes = [2, 3, 5, 7]

#we want to go item by item inside the list using for loop
for variable in primes: #variable will take the values of prime one by one
    print(variable)

2
3
5
7
```

In the above code

```
for i in primes
```

mean the variable `'i' iterates over all items in our list primes one by one.`

```
name = "Ethiopia" #this is our sequence
for i in name: #going one by one inside the string
    print(i)

E
t
h
i
o
p
i
a

for cherinet in oneToFive: #[1, 2 ,3, 4, 5]
  print(cherinet)

1
2
3
4
5
```

```python
TenToFifty = [i for i in range(10, 51)]
TenToFifty
```

```
[10,
 11,
 12,
 13,
 14,
 15,
 16,
 17,
 18,
 19,
 20,
 21,
 22,
 23,
 24,
 25,
 26,
 27,
 28,
 29,
 30,
 31,
 32,
 33,
 34,
 35,
 36,
 37,
 38,
 39,
 40,
 41,
 42,
 43,
 44,
 45,
 46,
 47,
 48,
 49,
 50]
```

```python
#take elements from TenToFifty list and multiply them by two

for x in TenToFifty: #[ 10, 11, 12.... 50]
  print(2*x)
```

```
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
52
54
56
58
60
62
64
66
68
70
72
74
76
78
80
82
84
86
88
90
92
94
96
98
100
```

```
for variable in sequence:
  do something
```

```
for x in TenToFifty: #[ 10, 11, 12.... 50]
  print(x + 10)
```

```
20
21
```

```
a = [1 ,2 ,3]
for i in a:
  print(i*2)
```

```
2
4
6
```

Range function

Range function. it can take 1, 2 or 3 arguments

when range has 1 argument(call it n), it is like looping over the list from 0 to n-1

syntax

```
range(number)
# this gives us list of numbers from 0 till number -1
#for example
range(10) #mean [0, 1, 2, 3,...9]

list(range(10))

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

list(range(7))

[0, 1, 2, 3, 4, 5, 6]

list(range(100))

[0,
 1,
 2,
 3,
 4,
 5,
 6,
 7,
 8,
 9,
 10,
 11,
 12,
 13,
 14,
 15,
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 94,
 95,
 96,
 97,
 98,
 99]
```

Range with 2 arguments

syntax

```
range(start, end)
#means from start to end-1
range(2, 6) #mean [2, 3, 4, 5]

list(range(2,7))

[2, 3, 4, 5, 6]

list(range(1,4))

[1, 2, 3]
```

Range with 3 arguments

syntax

```
range(start, end, skip)
# means from start to end-1 skipping some amount
range(2,9,3) # gives [2, 5, 8]

list(range(2,9,3)) #2, 5,8
```

```
[2, 5, 8]
```

```
list(range(1,9,2)) # 1, 3, 5, 7
```

```
[1, 3, 5, 7]
```

looping using index

```
for variable in range(value):
   do something
```

```
for i in range(5): # 0, 1, 2, 3, 4
    print('hello') #i = 4

#hello
#hello
#hello
#hello
#hello
```

```
for i in range(5): #i=0, 1, 2,3,4
    print('hello')
```

```
hello
hello
hello
hello
hello
```

```
for i in range(50): # 0, 1,...49.. 50 times
    print("Kebron")
```

```
Kebron
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```

```
for i in range(15): #i
  print(i)
```

```
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
```

```python
#let's print items in the list using index
primes = [2, 3, 5, 7, 11]
for i in range(5): #i values are [0, 1, 2, 3, 4]
    print(primes[i]) #primes[0]
```

```
2
3
5
7
11
```

```python
for i in primes:
  print(i)
```

```
2
3
5
7
11
```

```python
name = 'Ethiopia'
for i in range(len(name)): #range(8) 0, 1, 2...7
    print(name[i]) #name[0]
```

```python
name = 'Ethiopia'
for i in range(len(name)): #range(8).. 0, 1, 2, 3, .. 7
    print(name[i]) #name[0], name[1], name[2]
```

```
E
t
h
i
o
p
i
a
```

```python
l = [1, 'hello', 'name', 4]
# for j in range(4):  # [ 0, 1, 2, 3]
#     print(l[j]) #l[0], l[1], l[2], l[3]
for j in l:
    print(j)
# for j in l:
#     print(j)
```

```python
m=[1, 2, 3, 4]

for i in range(4): #0,1 ,2, 3
  print(m[i] * 2)
```

```
2
4
6
8
```

```
m=[1, 2, 3, 4, 5, 6, 7,8,9,10]

for i in range(1,5): #1, 2,3, 4,
  print(m[i] * 2)
```

```
4
6
8
10
```

```
m=[1, 2, 3, 4, 5, 6, 7,8,9,10]

for i in range(0,10,2): #0, 2, 4, 6...
  print(m[i] * 2)
```

```
2
6
10
14
18
```

```
m
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
for i in m:
  print(i * 2)
```

```
2
4
6
8
10
12
14
16
18
20
```

```
m
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

```
m=[1, 2, 3, 4]

for i in range(4): #0, 1, 2, 3
  m[i] = m[i] * 2 #m[0] = m[0]*2, m[0] = 1*2  [2, 2, 3, 4]
```

```
  #m[1] = m[1]*2, m[1] = 2*2 = 4.. [2, 4, 3, 4]
  #m[2] = m[2]*2, m[2] = 6, [2, 4, 6, 4]
  #m[3] = m[3]*2, m[3] = 8, [2, 4, 6, 8]
m
```

```
[2, 4, 6, 8]
```

```
l=[1, 2, 3, 4]
# output: l = [2, 4, 6, 8]
# l[0] = l[0]*2
# l[1] = l[1]*2
# l[2] = l[2]*2
# l[3] = l[3]*2

for i in range(4):
    l[i] = l[i]*2
l
```

```
[2, 4, 6, 8]
```

```
for x in range(2, 5): # numbers from 2 to 4
    print(x)

l = [1, 2, 3, 4, 5]
for i in range(len(l)):
    print(l[i])

for x in range(2, 8, 3): # numbers from 2 to 8 by skipping 3
    print(x)

# For loops can iterate over a sequence of numbers using the "range"
function

# Prints out the numbers 0,1,2,3,4
for x in range(5):
    print(x)


# Prints out 3,4,5
for x in range(3, 6):
    print(x)

# Prints out 3,5,7
for x in range(3, 8, 2):
    print(x)
```